



**ZIMBABWE SCHOOL EXAMINATIONS COUNCIL**  
**General Certificate of Education Advanced Level**

**PHYSICS**  
**PAPER 5**

**9188/5**

**NOVEMBER 2015 SESSION**

**1 hour 15 minutes**

Additional materials:

Answer paper

Electronic Calculator and / or Mathematical tables

Ruler (mm)

**TIME** 1 hour 15 minutes

**INSTRUCTIONS TO CANDIDATES**

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **four** questions.

**Question 1** is compulsory.

Answer any other **three** from the remaining questions.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

All working for numerical answers must be shown.

**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets [ ] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

Candidates are advised to spend 30 minutes on **question 1**.

**This question paper consists of 7 printed pages and 1 blank page.**

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## Formulae

uniformly accelerated motion,

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

work done on/by a gas,

$$W = p\Delta V$$

gravitational potential,

$$\phi = -\frac{Gm}{r}$$

refractive index,

$$n = \frac{1}{\sin C}$$

resistors in series,

$$R = R_1 + R_2 + \dots$$

resistors in parallel,

$$1/R = 1/R_1 + 1/R_2 + \dots$$

electric potential,

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

capacitors in series,

$$1/C = 1/C_1 + 1/C_2 + \dots$$

capacitors in parallel,

$$C = C_1 + C_2 + \dots$$

energy of charged capacitor,

$$W = \frac{1}{2}QV$$

alternating current/voltage,

$$x = x_0 \sin \omega t$$

hydrostatic pressure,

$$p = \rho gh$$

pressure of an ideal gas,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

radioactive decay,

$$x = x_0 \exp(-\lambda t)$$

decay constant,

$$\lambda = \frac{0.693}{t_{1/2}}$$

critical density of matter in the Universe,

$$\rho_0 = \frac{3H_0^2}{8\pi G}$$

equation of continuity,

$$Av = \text{constant}$$

Bernoulli equation (simplified),

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

Stokes' law,

$$F = 6\pi\eta r v$$

Reynolds' number,

$$R_e = \frac{\rho v r}{\eta}$$

drag force in turbulent flow,

$$F = Br^2\rho v^2$$



**Data**

|                               |                                                          |
|-------------------------------|----------------------------------------------------------|
| speed of light in free space, | $c = 3.00 \times 10^8 \text{ m s}^{-1}$                  |
| permeability of free space,   | $\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$           |
| permittivity of free space,   | $\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$     |
| elementary charge,            | $e = 1.60 \times 10^{-19} \text{ C}$                     |
| the Planck constant,          | $h = 6.63 \times 10^{-34} \text{ J s}$                   |
| unified atomic mass constant, | $u = 1.66 \times 10^{-27} \text{ kg}$                    |
| rest mass of electron,        | $m_e = 9.11 \times 10^{-31} \text{ kg}$                  |
| rest mass of proton,          | $m_p = 1.67 \times 10^{-27} \text{ kg}$                  |
| molar gas constant,           | $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$             |
| the Avogadro constant,        | $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$             |
| the Boltzmann constant,       | $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$              |
| gravitational constant,       | $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ |
| acceleration of free fall,    | $g = 9.81 \text{ m s}^{-2}$                              |



Answer question 1 and any other 3 from the remaining questions.

- 1 (a) (i) Define *tensile stress*.
- (ii) A vertical aluminium wire of length 1.6 m and cross-sectional area  $1.3 \times 10^{-6} \text{ m}^2$  carries a load of 60 N. The Young Modulus of aluminium is  $7.0 \times 10^{13} \text{ Pa}$ .

Calculate the

1. stress in the wire,
2. extension produced by the load.

[5]

- (b) (i) Define *e.m.f.*
- (ii) An electric bulb is marked '240 V, 100 W'.
1. Calculate the current in the bulb when operating normally.
  2. Describe **three** ways in which thermal energy can be transferred from the bulb filament when operating normally.

[6]

- (c) (i) Define *capacitance*.
- (ii) A capacitor is marked ' $100 \mu\text{F}$ , 12 V'.

Calculate the

1. energy,
2. charge,

stored by the capacitor when a p.d. of 12 V is applied across it.

[5]

- (d) (i) Explain the term *photon*.
- (ii) Estimate the energy of a red light photon.
- (iii) Deduce the region of the electromagnetic spectrum to which a photon of energy  $1 \times 10^{-25} \text{ J}$  belongs.

[5]



- (e) (i) Fig.1.1 shows the variation of pressure,  $P$ , with volume,  $V$ , of a gas.

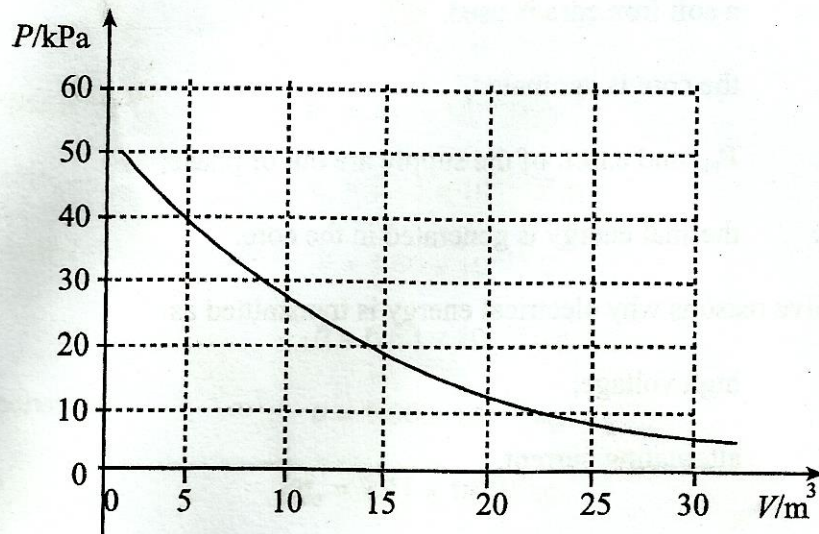


Fig.1.1

Calculate the work done by the gas as the volume increases from  $10 \text{ m}^3$  to  $30 \text{ m}^3$ .

- (ii) Suggest why a hot air balloon expands as it rises.

[3]

2

- (a) State Faraday and Lenz's laws of electromagnetic induction.

[2]

- (b) Fig.2.1 illustrates a transformer.

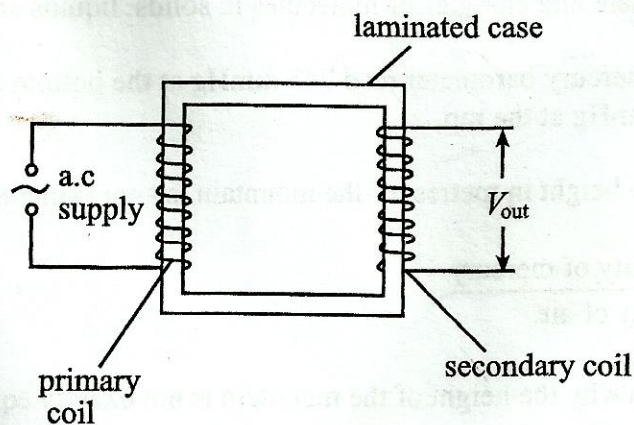


Fig.2.1

(i) Explain why

1. a soft iron core is used,
2. the core is laminated,
3.  $V_{\text{out}}$  and e.m.f. of the supply are out of phase,
4. thermal energy is generated in the core.

(ii) Give reasons why electrical energy is transmitted as

1. high voltage,
2. alternating current.

[8]

(c) Explain, using Lenz's law, why it is easier to switch on a bicycle dynamo when the lamps are switched off rather than when they are on.

[2]

3

(a) (i) Define *density*.

(ii) Explain why the densities of ice and water are almost equal but completely different to that of steam.

[3]

(b) Outline an experiment that demonstrates Brownian motion and state the evidence provided by this experiment.

[3]

(c) Compare the movement and energies of molecules in solids, liquids and gases.

[3]

(d) On a certain day, a mercury barometer read 745 mmHg at the bottom of a mountain and 590 mmHg at the top.

(i) Show that the height in metres, of the mountain is approximately equal to

$$\frac{0.158 \times \text{density of mercury}}{\text{density of air}}$$

(ii) Give a reason why the height of the mountain is not exactly equal to the one in (i).

[3]

4

(a) State any two factors which determine the choice of a thermometer for a particular measurement.

[2]

(b) Distinguish between an empirical scale of temperature and a thermodynamic scale of temperature.

[2]



- (c) Explain why different thermometers can only give the same temperature reading at fixed points. [2]
- (d) A resistance thermometer has a resistance of  $9.971 \Omega$  and  $14.043 \Omega$  at the ice and steam points respectively.

Determine the least change in temperature that can be detected if the minimum change in resistance is  $0.010 \Omega$ . [3]

- (e) A blast furnace worker is supposed to measure the temperature in the blast furnace which is about  $1\ 200^\circ\text{C}$ .

State, giving reasons, the thermometer that the worker can use. [3]

- 5 (a) (i) State any **two** sources of background radioactive radiation.

- (ii) **Table 5.1** shows properties of alpha and beta radiations.

Copy and complete **Table 5.1**.

**Table 5.1**

| radiation          | penetration                                                             | ionising ability | effects of electric field                               |
|--------------------|-------------------------------------------------------------------------|------------------|---------------------------------------------------------|
| alpha ( $\alpha$ ) |                                                                         |                  | deflected slightly in the same direction as +ve charge. |
| beta ( $\beta$ )   | many cm of air at s.t.p.<br>few mm of light metals<br>such as aluminium |                  | strong deflection in opposite direction to $\alpha$ .   |

- (iii) Explain why gamma rays are **not** used in paper thickness control during manufacturing. [7]
- (b) (i) Iodine-131 has a half-life of 8 days.
- Calculate the number of iodine-131 atoms remaining after 24 days for a sample with  $1.0 \times 10^{10}$  atoms.
- (ii) Explain how radiation can be harmful to a human being. [5]